

IS 1102/ SE 1102

Exception Handling

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Introduction to Exception Handling

- ▶ In a language without exception handling
 - ▶ When an exception occurs, control goes to the operating system, where a message is displayed and the program is terminated
- ▶ In a language with exception handling
 - ▶ Programs are allowed to trap some exceptions, thereby providing the possibility of fixing the problem and continuing

Basic Concepts

- Many languages allow programs to trap input/output errors
- An exception is any unusual event, either erroneous or not, detectable by either hardware or software, that may require special processing
- The special processing that may be required after detection of an exception is called exception handling
- The exception handling code unit is called an exception handler

Advantages of Built in Exception Handling

- Exception handling encourages programmers to consider many different possible errors
- Exception propagation allows a high level of reuse of exception handling code

Design Issues

- How and where are exception handlers specified and what is their scope?
- How is an exception occurrence bound to an exception handler?
- Can information about the exception be passed to the handler?
- Where does execution continue, if at all, after an exception handler completes its execution?

Design Issues

- Should there be default exception handlers for programs that do not provide their own?
- Are hardware detectable errors treated as exceptions that can be handled?
- How can exceptions be disabled, if at all?

Thank you